Advanced Control Systems

Assignment 2: Kinetic and Potential Energy Computation, CoM Positions, Partial Jacobians, and Inertia Matrix of a PRR Manipulator

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Abstract

This report presents the dynamic modeling and energy characterization of a 3-DOF PRR (Prismatic-Revolute-Revolute) robotic manipulator defined by the URDF specification. The study details the physical rationale and analytical derivation behind every dynamic modeling phase: link Center of Mass (CoM) position vectors, partial translational and rotational Jacobians ($\mathbf{J}_P^{l_i}, \mathbf{J}_O^{l_i}$), link local inertia tensors, the configuration-dependent generalized inertia matrix $\mathbf{B}(q)$, total kinetic energy $\mathcal{T}(q, \dot{q})$, and gravitational potential energy $\mathcal{U}(q)$. Every step is derived symbolically in parametric form and evaluated numerically using MATLAB.

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1 System Architecture & Dynamic Parameters

1.1 Description of the PRR Manipulator Dynamic Model

What is being done: We formulate the dynamic energy expressions for a spatial 3-degree-of-freedom PRR manipulator. The physical properties of the links are extracted directly from the URDF file (`00001_PRR_zzx FILIPPO SEGALLA.urdf`):

1. **Link 1 (Prismatic Joint, q_1):** A rectangular box (0.036 m $\times$ 0.036 m $\times$ 0.5 m) with mass $m_1 = 1.0$ kg.
2. **Link 2 (Revolute Joint, q_2):** A solid cylinder ($r = 0.019$ m, $h = 0.4$ m) aligned along Z_2 with mass $m_2 = 1.0$ kg.
3. **Link 3 (Revolute Joint, q_3):** A solid cylinder ($r = 0.019$ m, $h = 0.4$ m) aligned along X_3 with mass $m_3 = 1.0$ kg.

Why it is done: A complete dynamic description requires combining geometric kinematics with inertial properties (masses and inertia tensors). Establishing exact mass values and link geometry formulas is necessary to set up the Lagrangian $\mathcal{L} = \mathcal{T} - \mathcal{U}$, which governs the physical equations of motion.

1.2 Local Inertia Tensors ($\mathbf{I}_{l_i}^i$)

What is being done: For each link, the central inertia tensor $\mathbf{I}_{l_i}^i$ is evaluated with respect to a frame centered at its Center of Mass (CoM) and aligned with the link's principal axes of inertia:

- **Link 1 (Parallelepiped Box):**

$$I_{1xx} = I_{1yy} = \frac{1}{12} m_1 (0.036^2 + 0.5^2), \quad I_{1zz} = \frac{1}{12} m_1 (0.036^2 + 0.036^2) \tag{1}$$

- **Link 2 (Cylinder along Z_2):**

$$I_{2xx} = I_{2yy} = \frac{1}{12} m_2 (3 \cdot 0.019^2 + 0.4^2), \quad I_{2zz} = \frac{1}{2} m_2 (0.019^2) \tag{2}$$

- **Link 3 (Cylinder along X_3):**

$$I_{3xx} = \frac{1}{2} m_3 (0.019^2), \quad I_{3yy} = I_{3zz} = \frac{1}{12} m_3 (3 \cdot 0.019^2 + 0.4^2) \tag{3}$$

Why it is done: Inertia tensors quantify a body's resistance to rotational acceleration. By expressing inertia tensors in local central frames aligned with the principal axes, off-diagonal cross-products vanish ($\mathbf{I}_{l_i}^i = \text{diag}(I_{ixx}, I_{iyy}, I_{izz})$). When the robot moves, these constant local tensors are transformed into the global frame using rotation matrices: $\mathbf{I}_{l_i}(q) = \mathbf{R}_i(q) \mathbf{I}_{l_i}^i \mathbf{R}_i^T(q)$.

2 Centers of Mass & Partial Jacobians

2.1 Centers of Mass (CoM) Positions in Base Frame

What is being done: We locate the position vector $\mathbf{p}_{l_i}$ of each link's CoM relative to the global base frame Σ_0 using cumulative homogeneous transformation matrices:

$$\begin{bmatrix} \mathbf{p}_{l_i} \\ 1 \end{bmatrix} = \mathbf{T}_i^0(q) \begin{bmatrix} \mathbf{p}_{l_i}^i \\ 1 \end{bmatrix} \tag{4}$$

Where the local CoM vectors relative to the D-H origins are:

$$\mathbf{p}_{l_1}^1 = \begin{bmatrix} 0 \\ 0 \\ -0.25 \\ 1 \end{bmatrix}, \quad \mathbf{p}_{l_2}^2 = \begin{bmatrix} 0 \\ 0 \\ -0.20 \\ 1 \end{bmatrix}, \quad \mathbf{p}_{l_3}^3 = \begin{bmatrix} -0.20 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{5}$$

Evaluating the transformations symbolically yields:

$$\mathbf{p}_{l_1}(q) = \begin{bmatrix} 0 \\ 0 \\ q_1 + 0.40 \end{bmatrix}, \quad \mathbf{p}_{l_2}(q) = \begin{bmatrix} 0 \\ 0 \\ q_1 + 0.85 \end{bmatrix}, \quad \mathbf{p}_{l_3}(q) = \begin{bmatrix} -0.20 \sin q_2 \cos q_3 \\ 0.20 \cos q_2 \cos q_3 \\ q_1 + 0.20 \sin q_3 + 1.05 \end{bmatrix} \tag{6}$$

Why it is done: König's theorem states that the total kinetic energy of a rigid link is the sum of translational energy at its CoM and rotational energy about its CoM. Knowing $\mathbf{p}_{l_i}(q)$ is necessary for calculating both the linear velocities of the link centers of mass and the gravitational potential energy $\mathcal{U}_i = -m_i \mathbf{g}_0^T \mathbf{p}_{l_i}$.

2.2 Partial Jacobians ($\mathbf{J}_P^{l_i}, \mathbf{J}_O^{l_i}$)

What is being done: Partial Jacobians map generalized joint velocities $\dot{q}$ to the linear velocity $\dot{\mathbf{p}}_{l_i}$ and angular velocity $\boldsymbol{\omega}_i$ of link i:

$$\dot{\mathbf{p}}_{l_i} = \mathbf{J}_P^{l_i}(q)\dot{q}, \quad \boldsymbol{\omega}_i = \mathbf{J}_O^{l_i}(q)\dot{q} \tag{7}$$

The columns j ($j \leq i$) of the partial Jacobians are computed as:

- **Prismatic Joint j:** $\mathbf{J}_{Pj}^{l_i} = \mathbf{z}_{j-1}$, $\mathbf{J}_{Oj}^{l_i} = \mathbf{0}$.
- **Revolute Joint j:** $\mathbf{J}_{Pj}^{l_i} = \mathbf{z}_{j-1} \times (\mathbf{p}_{l_i} - \mathbf{p}_{j-1})$, $\mathbf{J}_{Oj}^{l_i} = \mathbf{z}_{j-1}$.
- **Inactive Joints ($j > i$):** $\mathbf{J}_{Pj}^{l_i} = \mathbf{0}$, $\mathbf{J}_{Oj}^{l_i} = \mathbf{0}$.

Why it is done: Since link i is affected only by joints $1, \ldots, i$, columns beyond i are zero. Partial Jacobians allow us to write the kinetic energy of each link in a compact quadratic form without explicitly dealing with velocity vector products.

3 Generalized Inertia Matrix $\mathrm{B}(q)$

3.1 Mathematical Formulation

What is being done: The generalized inertia matrix $\mathbf{B}(q) \in \mathbb{R}^{3\times3}$ is constructed by summing the translational and rotational kinetic energy contributions of all links:

$$\mathbf{B}(q) = \sum_{i=1}^{3} \left(m_i (\mathbf{J}_P^{l_i})^T \mathbf{J}_P^{l_i} + (\mathbf{J}_O^{l_i})^T \mathbf{R}_i \mathbf{I}_{l_i}^i \mathbf{R}_i^T \mathbf{J}_O^{l_i} \right) \tag{8}$$

Evaluating this sum symbolically yields the parametric expression:

$$\mathbf{B}(q) = \begin{bmatrix} m_1 + m_2 + m_3 & 0 & \frac{m_3}{5}\cos q_3 \\ 0 & I_{2yy} + I_{3yy}\cos^2 q_3 + I_{3xx}\sin^2 q_3 + \frac{m_3}{25}\cos^2 q_3 & 0 \\ \frac{m_3}{5}\cos q_3 & 0 & I_{3zz} + \frac{m_3}{25} \end{bmatrix} \tag{9}$$

Substituting numerical parameters gives:

$$\mathbf{B}(q) = \begin{bmatrix} 3.0 & 0 & 0.2\cos q_3 \\ 0 & 0.1068 - 0.05324\sin^2 q_3 & 0 \\ 0.2\cos q_3 & 0 & 0.05342 \end{bmatrix} \tag{10}$$

Why it is done & Physical Properties:

1. **Symmetry ($\mathbf{B} = \mathbf{B}^T$):** Reflects the reciprocal dynamic coupling between joints.
2. **Positive Definiteness ($\mathbf{B} > 0$):** Ensures kinetic energy is strictly positive for non-zero velocities ($\mathcal{T} > 0$ for $\dot{q} \neq 0$).
3. **Coupling Term ($B_{13} = 0.2\cos q_3$):** Quantifies the inertial interaction between vertical base translation (q_1) and wrist rotation (q_3).
4. **Invariance to q_2:** $\mathbf{B}(q)$ is independent of q_2 because rotation about the vertical Z-axis does not alter the manipulator's mass distribution relative to gravity or rotation axes.

4 Kinetic & Potential Energy Computation

4.1 Total Kinetic Energy $\mathcal{T}(q, \dot{q})$

What is being done: The total kinetic energy is formulated as a quadratic expression in joint velocities:

$$\mathcal{T}(q, \dot{q}) = \frac{1}{2} \dot{q}^T \mathbf{B}(q) \dot{q} \tag{11}$$

Evaluating the matrix product numerically yields:

$$\mathcal{T}(q, \dot{q}) = 1.5\dot{q}_1^2 + 0.2\dot{q}_1\dot{q}_3\cos q_3 - 0.02662\dot{q}_2^2\sin^2 q_3 + 0.05342\dot{q}_2^2 + 0.02671\dot{q}_3^2 \tag{12}$$

Why it is done: Kinetic energy stores the mechanical work done by joint actuators during motion. The quadratic term $\frac{1}{2}B_{ii}\dot{q}_i^2$ represents the effective inertia at joint i, while cross-terms $B_{ij}\dot{q}_i\dot{q}_j$ represent dynamic cross-coupling.

4.2 Total Potential Energy $\mathcal{U}(q)$

What is being done: Potential energy due to gravity ($\mathbf{g}_0 = [0, 0, -g]^T$) is calculated as:

$$\mathcal{U}(q) = -\sum_{i=1}^{3} m_i \mathbf{g}_0^T \mathbf{p}_{l_i}(q) \tag{13}$$

Evaluating symbolically gives:

$$\mathcal{U}(q) = gm_1\left(q_1 + \frac{2}{5}\right) + gm_2\left(q_1 + \frac{17}{20}\right) + gm_3\left(q_1 + \frac{\sin q_3}{5} + \frac{21}{20}\right) \tag{14}$$

Substituting numerical values ($g = 9.81$ m/s^2, $m_i = 1$ kg):

$$\mathcal{U}(q) = 29.43q_1 + 1.962\sin q_3 + 22.56 \tag{15}$$

Why it is done: Potential energy depends strictly on position coordinates. For this PRR robot, $\mathcal{U}(q)$ depends on q_1 (vertical translation) and q_3 (elevation of Link 3), but is completely independent of q_2, since rotation about Z_0 moves the robot within a horizontal plane perpendicular to gravity.